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Neuro-Symbolic AI vs. Vector Symbolic Architectures

Learning, architecture and reasoning

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Where does symbolic structure live?

SYSTEMS QUESTION

A reasoning system must ground symbols, compose them, and select an answer.

NSAI

Learned embeddings feed an explicit reasoning module.

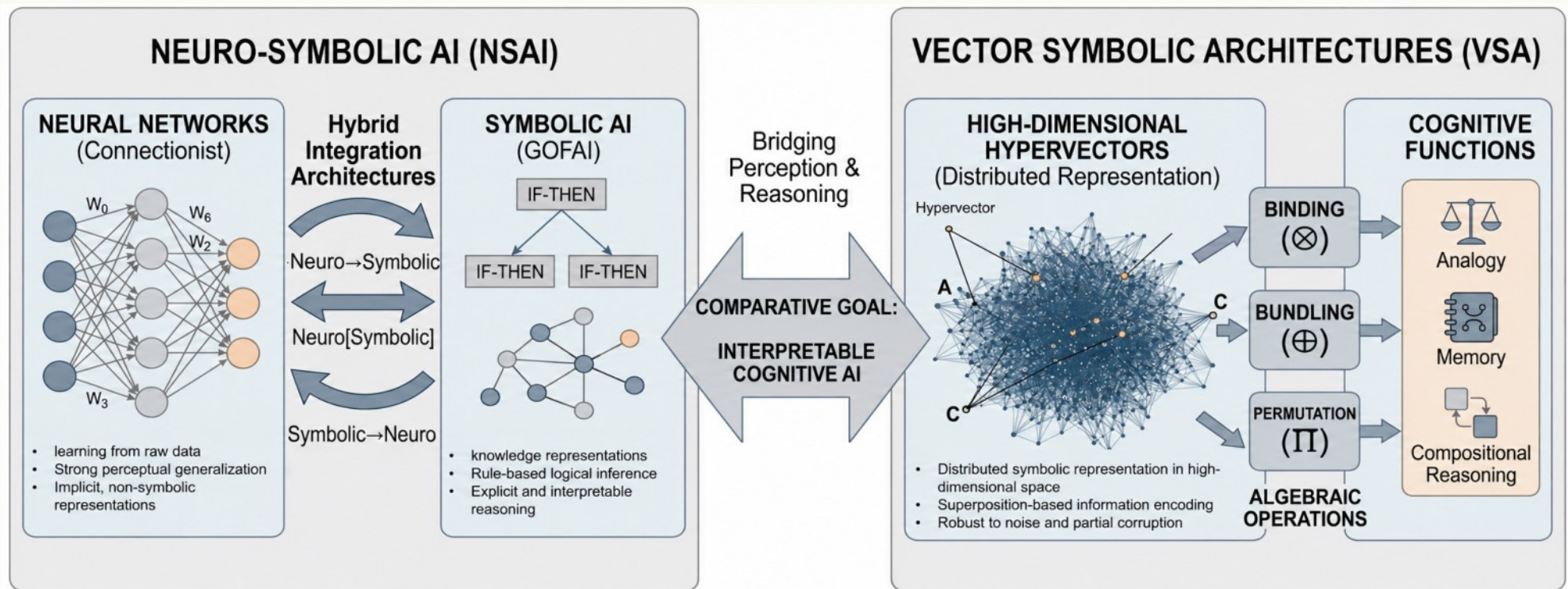
VSA / HDC

Hypervector operations carry structure within a shared vector space.

We compare representation, learning, architecture, and reasoning accuracy.

A common perception-to-reasoning goal

The two paradigms place compositional structure in different parts of the system.



NSAI makes symbolic structure explicit. VSA stores compositional structure in vector geometry.

NSAI: learned symbols and structured reasoning

REPRESENTATION

A neural encoder turns input into object or relation embeddings.

REASONING

A structured module repeatedly transforms symbolic states.

LEARNING

Task supervision can be combined with structural or logical constraints.

$$\mathbf{L} = \mathbf{L}_{\text{task}} + \lambda \mathbf{L}_{\text{structure}}$$

VSA: composition in hypervector space

HIGH-DIMENSIONAL HYPERVECTORS

Symbols map to distributed vectors, often with 10,000 dimensions in our experiment.

THREE CORE OPERATIONS

Binding combines roles with values

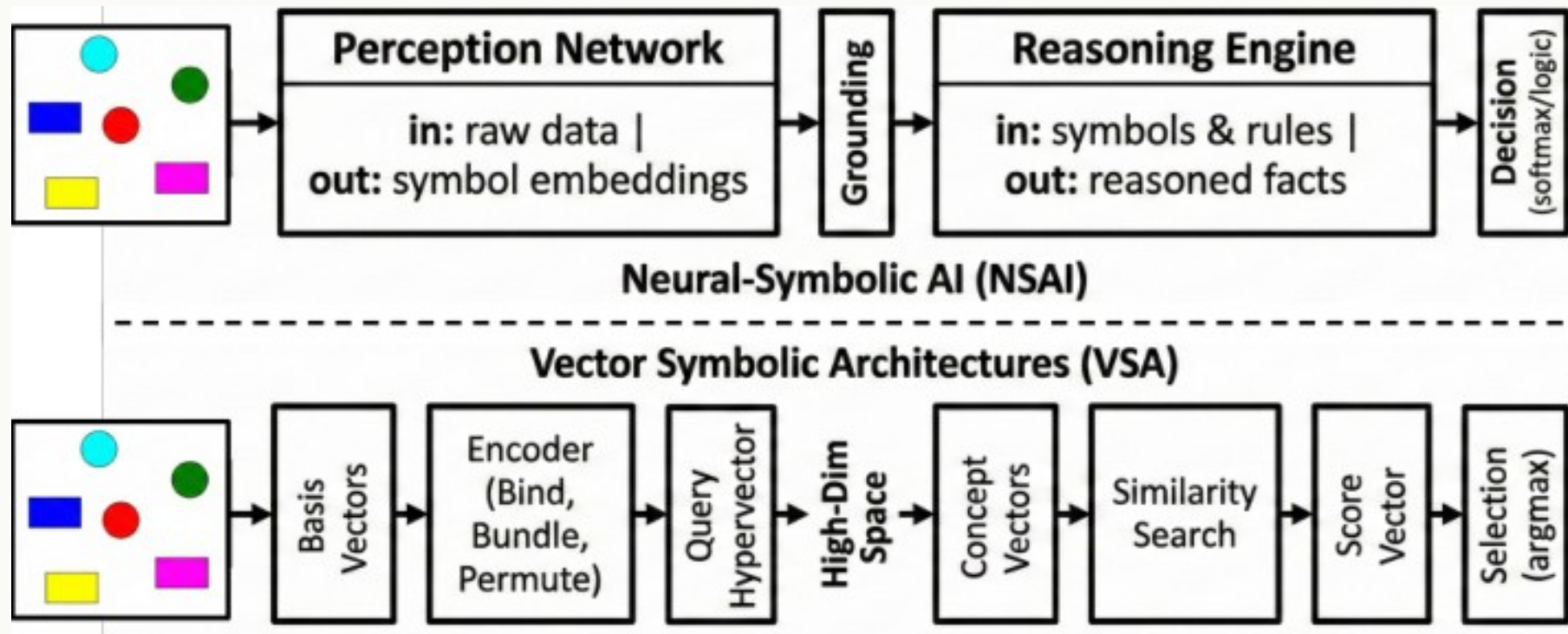
Bundling superposes related items

Permutation encodes order or position

Prediction: compare the query vector with class prototypes in associative memory.

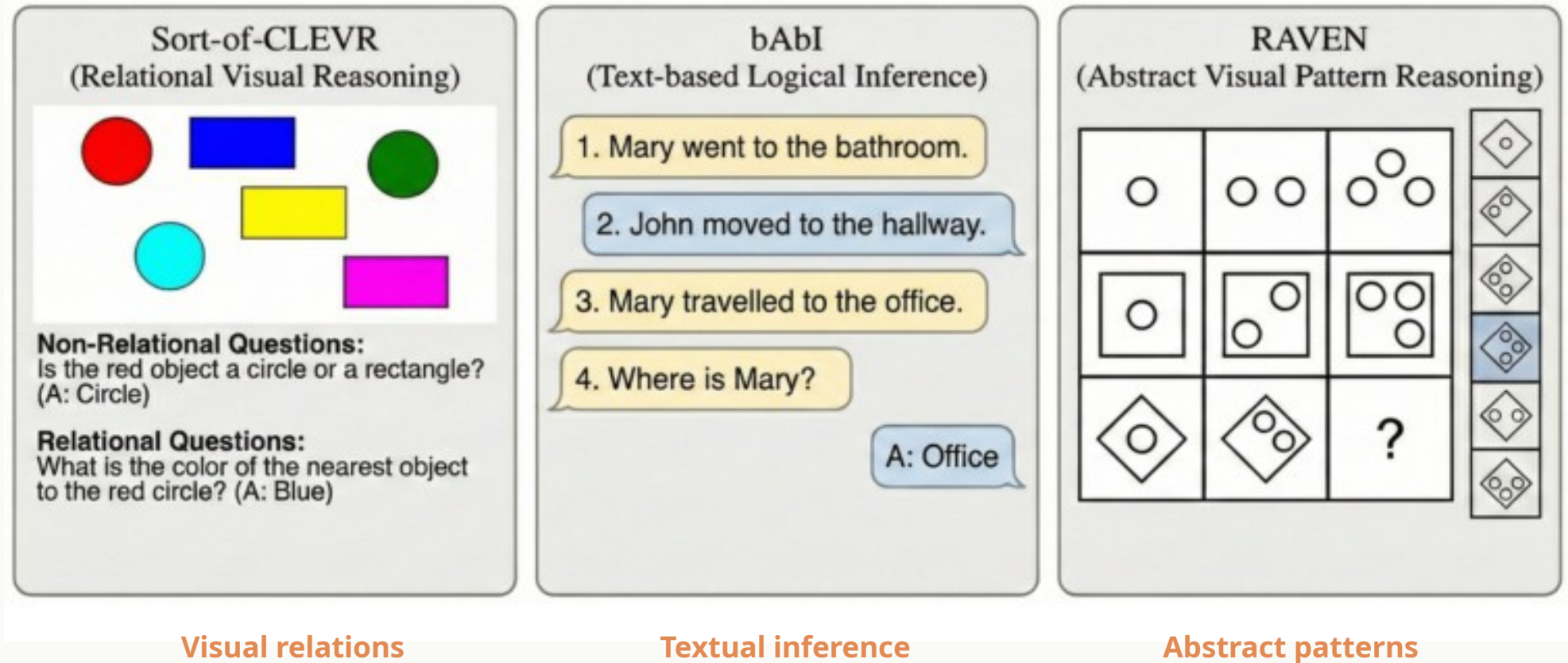
Architecture: the decision paths

Sort-of-CLEVR provides a common visual input for the two conceptual pipelines.



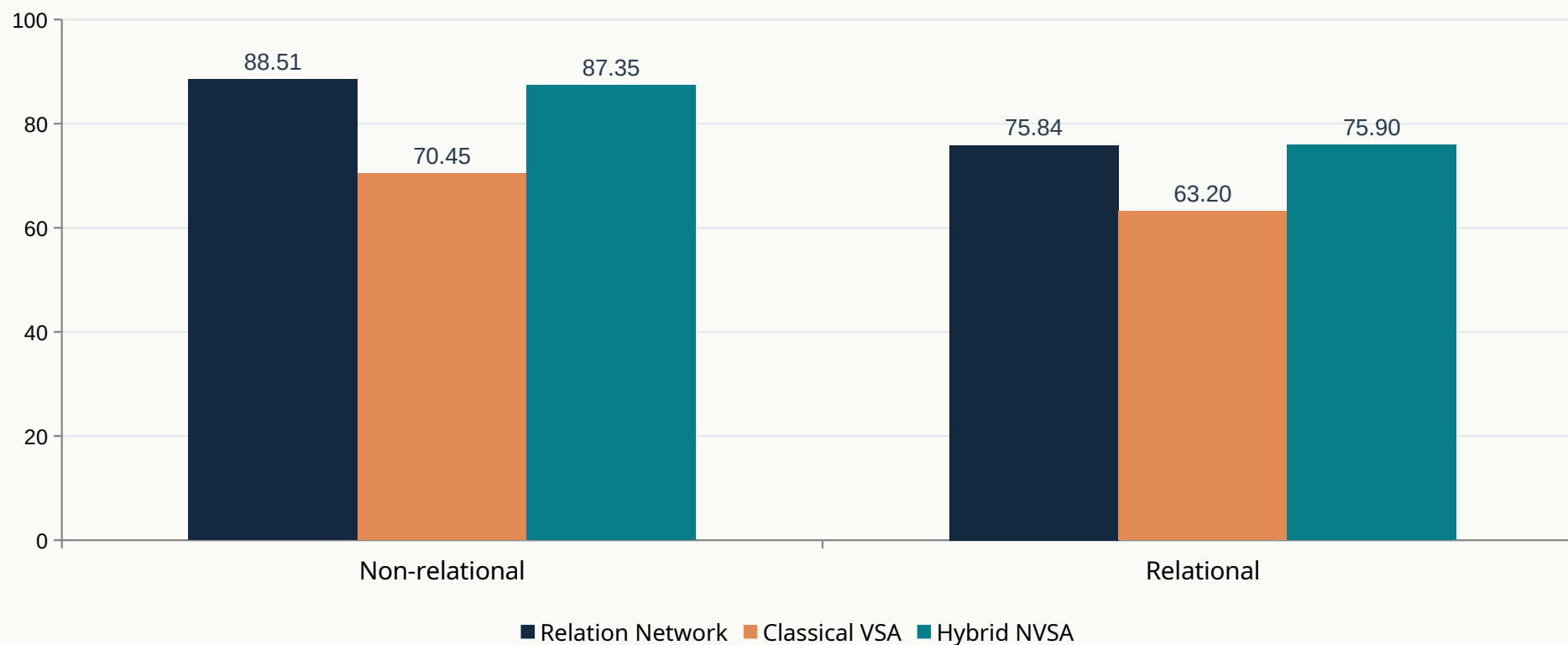
NSAI: learned perception and explicit reasoning **VSA: encoding and similarity search**

Evaluation spans three reasoning demands



Sort-of-CLEVR: hybrid narrows the gap

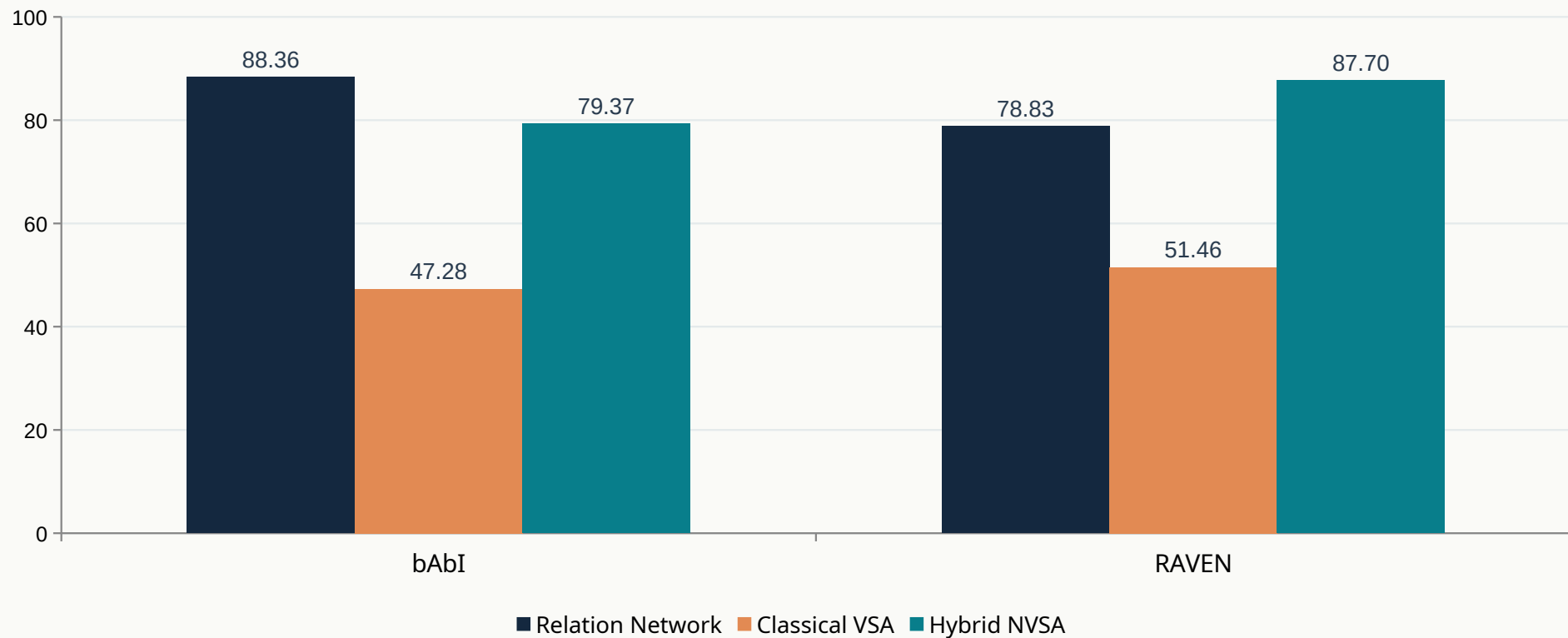
Accuracy (%) from Table I; selected models shown



NVSA reaches 87.35% and 75.90%, near or above RN in these two settings.

bAbI and RAVEN favor different designs

Accuracy (%) from Table I; selected models shown



RN leads on bAbI; NVSA leads the reported comparison on RAVEN.

Architectural implications and limits

NSAI

Learned, task-specific symbolic embeddings

Explicit reasoning modules and gradient-based training

VSA

Distributed, algebraically composed hypervectors

Implicit composition and prototype-based updates

Evidence: accuracy across three benchmarks. Next: controlled ablations and system costs.

What to remember

- 1 Both paradigms connect perception with structured reasoning.
- 2 They place symbolic structure in different parts of the system.
- 3 In the reported benchmarks, NVSA improves over classical VSA and leads on RAVEN.

Thank you. Questions?

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